

Automated Angiography Blood Vessel Segmentation and Morphological Analysis using Deep Learning Architectures

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By: Seemant

Supervisor: Dr. Predeep Paik

Indian Institute of Technology (BHU)
Varanasi, India

Chapter 1: Introduction

Introduction and Clinical Motivation

Cardiovascular diseases (CVDs) and systemic vascular disorders remain the leading causes of global mortality, necessitating continuous advancements in diagnostic imaging and preventative clinical care. The structural and morphological integrity of the human vascular network serves as a direct biomarker for a multitude of severe pathological conditions, including hypertension, arteriosclerosis, diabetic retinopathy, and various forms of ischemia. In clinical practice, angiography---a specialized medical imaging technique utilized to visualize the inside, or lumen, of blood vessels and organs of the body---has become the gold standard for diagnosing vascular abnormalities. Through the injection of radio-opaque contrast agents and the use of X-ray imaging, fluoroscopy, or advanced optical coherence tomography (OCT), angiographic imaging provides physicians with a high-resolution map of the circulatory system.

Despite the high fidelity of modern angiographic imaging modalities, the extraction of actionable quantitative metrics from these images remains a significant bottleneck. The precise segmentation and topological extraction of blood vessels from angiography images are of paramount importance. A segmented vascular map allows for the calculation of critical morphological features such as vessel diameter, tortuosity, branching angles, and the total count of distinct vascular networks. These metrics are indispensable for the early detection of microvascular abnormalities, the planning of intricate surgical interventions, such as stent placement and bypass surgeries, and the longitudinal monitoring of disease progression.

Traditionally, the segmentation of blood vessels has been performed manually by highly trained clinicians and radiologists. However, manual annotation is a fundamentally flawed approach in the context of modern, high-throughput clinical environments. It is exceedingly labor-intensive, taking hours to precisely annotate a single high-resolution angiogram. Furthermore, manual segmentation is highly subjective and prone to inter-observer and intra-observer variability. The cognitive fatigue associated with tracing complex, tortuous, and often overlapping vascular structures inevitably leads to human error. Consequently, the development of fully automated, highly precise, and computationally efficient methods for blood vessel segmentation is not merely an academic pursuit

but an urgent clinical necessity.

Challenges in Automated Blood Vessel Segmentation

The transition from manual annotation to automated algorithmic segmentation is fraught with formidable technical challenges. While early attempts utilizing traditional image processing techniques---such as matched filtering, Hessian-based Frangi filters, and morphological operations---showed initial promise, they fundamentally struggle to generalize across diverse clinical datasets. The difficulties in achieving robust automated segmentation stem from several factors inherent to both the biological complexity of the vascular system and the physics of the image acquisition process:

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High Structural Complexity and Variance

The human vascular network is an incredibly dense and topologically complex fractal structure. Vessels exhibit extreme variance in scale, ranging from prominent, thick primary arteries and veins to exceptionally thin terminal micro-capillaries that may be only one or two pixels wide in digital representations. Furthermore, vessels branch at varying angles, overlap in two-dimensional projections, and display high degrees of tortuosity. An automated system must possess the representational capacity to simultaneously recognize large, high-contrast structures and preserve the delicate continuity of ultra-thin micro-vessels. Failure to do so results in fragmented segmentations that are useless for downstream topological analysis, such as skeletonization.

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Severe Class Imbalance

In the context of binary semantic segmentation, angiographic images present a severe class imbalance problem. The pixels belonging to the foreground class (blood vessels) typically constitute only a small fraction---often less than 10%---of the total image area, while the vast majority of pixels belong to the background class (tissue, retina, or surrounding anatomical structures). Standard loss functions, such as standard Cross-Entropy, heavily penalize the model based on pixel volume. Consequently, a naive deep learning model trained on such imbalanced data will often collapse into a degenerate state, predicting the entire image as background to achieve an artificially high, yet

clinically useless, global accuracy. Specialized loss formulations are absolutely necessary to force the optimization landscape to focus on the sparse foreground structures.

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Low Contrast and Uneven Illumination

The acquisition of angiographic images is highly susceptible to physical artifacts. Uneven distribution of the contrast agent, variations in tissue thickness, and the inherent physics of X-ray or optical scattering lead to non-uniform background illumination and localized regions of exceedingly low contrast. Micro-capillaries, in particular, often fade into the background noise, possessing pixel intensities that are statistically indistinguishable from surrounding tissue artifacts. Furthermore, central light reflections and shadowing effects can create false edges that mislead traditional gradient-based edge detection algorithms.

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Pathological Interferences and Artifacts

Clinical angiograms are rarely pristine. The very pathologies that necessitate the imaging often introduce severe visual interferences. The presence of lesions, exudates, hemorrhages, and surgical artifacts (such as previously placed stents or clips) introduces significant noise. These pathological structures can visually obscure the underlying vessels or, conversely, mimic their tubular appearance, confusing both traditional algorithms and poorly regularized machine learning models.

The Paradigm Shift to Deep Learning

To overcome the limitations of traditional, hand-crafted feature engineering, the field of medical image analysis has witnessed a paradigm shift towards Deep Learning (DL), specifically Convolutional Neural Networks (CNNs). Unlike traditional methods that rely on heuristic rules, deep learning models autonomously discover hierarchical, data-driven feature representations directly from annotated clinical datasets.

The introduction of Fully Convolutional Networks (FCNs) marked the beginning of modern semantic segmentation, allowing networks to output dense, pixel-wise classifications rather than single

image-level labels. This was rapidly superseded by the U-Net architecture, which introduced a symmetric encoder-decoder structure with skip connections. These skip connections bypassed the spatial bottleneck, fusing high-resolution, low-level spatial features from the encoder with low-resolution, high-level semantic features from the decoder. U-Net quickly became the de facto standard for medical image segmentation.

However, as clinical demands for precision increased, particularly for the extraction of ultra-thin vessels, the limitations of the standard U-Net became apparent. The semantic gap between the encoder and decoder feature maps often results in the loss of fine-grained spatial details, leading to disconnected vessel segments. This thesis explores advanced architectures designed to bridge this semantic gap and capture multi-scale contextual information more effectively. Specifically, we investigate UNet++, an architecture featuring nested, dense skip pathways, and DeepLabV3+, an architecture utilizing Atrous Spatial Pyramid Pooling (ASPP) to capture features at multiple receptive field scales without degrading spatial resolution.

Research Objectives and Questions

The overarching goal of this thesis is to engineer a highly accurate, automated, end-to-end pipeline for the semantic segmentation and subsequent morphological analysis of blood vessels from angiography images. To achieve this, the research is guided by the following specific objectives and research questions:

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Primary Objectives

- Implementation of Advanced Architectures: To design, implement, and optimize two state-of-the-art deep learning architectures---UNet++ (with and without a pre-trained ResNet50 backbone) and DeepLabV3+---for the specific task of binary vessel segmentation.

- Mitigation of Class Imbalance: To formulate, apply, and evaluate advanced hybrid loss functions, specifically combining Binary Cross-Entropy (BCE) with Dice Loss, and Focal Loss with Dice Loss, to counteract the severe foreground-background pixel imbalance inherent in angiograms.

- Morphological Feature Extraction: To develop a robust post-processing pipeline utilizing mathematical morphology and skeletonization (thinning) algorithms to extract the centerline topology of the predicted vessel masks, enabling the automated counting of distinct connected vessel

networks, branch points, and endpoints.

- **Rigorous Empirical Evaluation:** To systematically evaluate the proposed methodologies on a substantial dataset of 637 annotated angiographic images, comparing the architectures across standard clinical metrics including Intersection over Union (IoU), Dice Coefficient, Precision, and Recall.

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Key Research Questions

- **RQ1:** How does the dense, nested skip pathway architecture of UNet++ compare to the Atrous Spatial Pyramid Pooling mechanism of DeepLabV3+ in preserving the spatial continuity of ultra-thin micro-capillaries in noisy angiograms?

- **RQ2:** To what extent do hybrid loss formulations (BCE+Dice vs. Focal+Dice) accelerate convergence and improve the absolute Intersection over Union (IoU) compared to standard loss functions when dealing with highly imbalanced vascular datasets?

- **RQ3:** Can downstream mathematical skeletonization be reliably applied to the probabilistic outputs of deep learning models to extract clinically meaningful, quantitative morphological biomarkers, such as the total count of distinct vessel networks?

Thesis Contributions

This thesis makes several substantial contributions to the fields of medical image analysis and computer vision. By combining state-of-the-art semantic segmentation architectures with rigorous topological post-processing, this work bridges the gap between raw pixel classification and clinically actionable quantitative metrics. The major contributions are enumerated below:

- **Comprehensive Architectural Comparison:** We provide a rigorous, head-to-head empirical evaluation of UNet++ and DeepLabV3+ in the specific domain of angiography. We demonstrate how the nested decoder of UNet++ and the ASPP module of DeepLabV3+ uniquely handle the multi-scale challenges of vessel extraction, providing valuable insights into architectural selection for tubular structure segmentation.

- **Optimized Hybrid Loss Formulations:** We successfully implement and validate hybrid loss functions that force the gradient descent optimization to prioritize overlapping regional predictions (Dice) while simultaneously penalizing hard-to-classify pixels (Focal Loss). We demonstrate that

these formulations prevent background collapse and significantly elevate the Dice Coefficient on the validation set.

- Integration of Heavy Data Augmentation: We integrate the Albumentations library to apply severe, dynamic spatial and pixel-level augmentations during training. We show that this approach drastically reduces overfitting and improves the generalization of the models on unseen test distributions.

- Automated Topological Biomarker Extraction: Moving beyond simple pixel accuracy, we introduce a complete downstream pipeline that utilizes iterative skeletonization algorithms on the predicted masks. This allows the system to autonomously count the number of distinct, disconnected vascular networks and map branch/endpoints, transitioning the output from a visual mask to a quantitative clinical report.

Organization of the Thesis

To provide a clear, logical, and rigorous exposition of the research conducted, the remainder of this thesis is structured as follows:

Chapter 2: Literature Review provides an extensive, chronological review of the state-of-the-art in medical image segmentation. It covers the evolution of algorithms from early hand-crafted filters to the dominance of Convolutional Neural Networks, detailing the architectural innovations of the U-Net family and the DeepLab series. It also reviews the literature on loss functions for imbalanced data and post-processing skeletonization techniques.

Chapter 3: Preliminaries and System Model lays the mathematical and theoretical foundation for the research. It defines the formal representation of the image data, the mathematical formulation of the geometric and pixel-level augmentations utilized, the precise derivations of the evaluation metrics (IoU, Dice, Precision, Recall), and the mathematical definitions of the applied loss functions.

Chapter 4: Proposed Methodology provides a deep, block-by-block architectural breakdown of the implemented systems. It details the dense skip pathways and ResNet50 encoder of the UNet++ models, the Atrous Spatial Pyramid Pooling mechanics of DeepLabV3+, the training pipeline algorithms, and the specific iterative algorithms employed for post-segmentation morphological skeletonization.

Chapter 5: Experimental Design outlines the rigorous experimental setup. It details the composition and statistical distribution of the 637-image dataset (500 train, 137 test), the specific hyperparameter

configurations, the learning rate scheduling strategies, and the hardware/software ecosystem utilized to execute the experiments.

Chapter 6: Results and Discussion presents the empirical findings of the research. It includes extensive comparative tables of quantitative metrics, plots of the training and validation learning curves, and qualitative visual analyses of the predicted segmentation overlays. Crucially, it discusses the outputs of the skeletonization algorithm, analyzing the distinct vessel counts extracted from the test set.

Finally, Chapter 7: Conclusion and Future Work summarizes the core findings and contributions of the thesis, acknowledges current limitations (such as the handling of severely fragmented vessels), and proposes specific, actionable avenues for future research, including the transition to 3D volumetric segmentation and real-time clinical deployment.

Chapter 2: Literature Review

The task of automated blood vessel segmentation from medical imagery, specifically angiography and fundus photography, has been the subject of extensive academic research for several decades. The structural extraction of the vascular network is widely recognized as a critical preliminary step in the computerized analysis of a vast array of pathologies. This chapter provides a comprehensive, chronological review of the methodologies developed for blood vessel segmentation, tracing the evolution from early hand-crafted heuristic filters to the modern era of deep convolutional neural networks. Furthermore, it extensively reviews the literature surrounding the architectural innovations of the UNet and DeepLab families, the mathematical formulations designed to combat severe class imbalance, and the classical algorithms utilized for post-segmentation morphological skeletonization.

Traditional and Hand-crafted Segmentation Techniques

Prior to the proliferation of deep learning, vessel segmentation relied heavily on explicit, hand-crafted feature engineering based on the known geometric and photometric properties of blood vessels. These early methods generally assumed that vessels appear as continuous, piecewise-linear tubular structures that are darker (or lighter, depending on the imaging modality) than the surrounding tissue.

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Matched Filtering and Edge Detection

One of the earliest and most widely adopted techniques was the matched filter response. [textcite{fraz2012blood}](#) in their comprehensive survey noted that matched filters operate by convolving the image with multiple 2D Gaussian kernels at varying orientations. The parameters of the Gaussian curves are explicitly tuned to match the expected width profile of the blood vessels. While computationally efficient, matched filters suffer from high false-positive rates when confronted with non-vascular structures that exhibit similar contrast profiles, such as hemorrhages, exudates, or the boundaries of the optic disc.

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Hessian-based and Frangi Vesselness Filters

To overcome the limitations of simple matched filters, researchers turned to differential geometry and multi-scale analysis. The most influential work in this domain was the formulation of the vesselness measure by Frangi et al. The Frangi filter computes the eigenvalues of the Hessian matrix at multiple spatial scales across the image. By analyzing the ratio of the eigenvalues, the filter can distinguish between blob-like structures (e.g., aneurysms), plate-like structures, and the desired tubular structures (vessels). While the Frangi filter and its numerous derivatives significantly improved the continuity of extracted vessels, they remain highly sensitive to noise and require extensive hyperparameter tuning for different datasets and imaging modalities.

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Morphological Operations

Mathematical morphology has also played a crucial role in traditional segmentation pipelines. Operations such as the morphological top-hat transformation are frequently used to enhance vessel contrast and correct for the non-uniform background illumination inherent in clinical imaging. However, morphological methods alone are rarely sufficient for final segmentation; they are typically employed as preprocessing steps to enhance the performance of subsequent thresholding or region-growing algorithms.

The fundamental limitation of all traditional, hand-crafted methods is their inability to generalize. A filter explicitly tuned for the thick primary arteries of the optic disc will inevitably fail to detect the ultra-thin micro-capillaries in the periphery, and vice versa. The reliance on rigid heuristic rules makes these systems highly brittle in the presence of noise, artifacts, and pathological deviations.

The Advent of Deep Learning in Semantic Segmentation

The paradigm of computer vision underwent a seismic shift with the advent of deep Convolutional Neural Networks (CNNs). Unlike traditional methods, CNNs do not rely on human-engineered features; instead, they learn a hierarchy of features directly from raw pixel data through backpropagation.

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Fully Convolutional Networks (FCN)

The breakthrough for pixel-level semantic segmentation came with the introduction of the Fully Convolutional Network (FCN) by [Long et al. \(2015\)](#). Prior to FCN, CNNs were primarily used for image-level classification, utilizing fully connected layers that collapsed spatial dimensions into a single vector. Long et al. demonstrated that by replacing these fully connected layers with convolutional layers, the network could output a spatial map of class probabilities. Furthermore, FCN introduced the concept of transposed convolutions (or deconvolutions) to up-sample the low-resolution feature maps back to the original image dimensions. While groundbreaking, the original FCN architecture produced coarse, blob-like segmentations because the up-sampling process fundamentally struggled to recover the fine-grained spatial details lost during the successive pooling operations of the encoder.

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The U-Net Revolution

The limitations of the FCN architecture in the medical domain were decisively addressed by [Ronneberger et al. \(2015\)](#) with the introduction of the U-Net architecture. Originally developed for the segmentation of neuronal structures in electron microscopic stacks, U-Net established a symmetric, U-shaped encoder-decoder topology. The critical innovation of U-Net was the introduction of skip connections. These connections directly concatenated the high-resolution, low-level spatial feature maps from the contracting path (encoder) with the corresponding low-resolution, high-level semantic feature maps in the expanding path (decoder).

This structural fusion allowed the decoder to utilize both the abstract semantic context ("what is this object?") and the precise spatial localization ("where are its exact boundaries?"). U-Net rapidly became the de facto standard for medical image segmentation due to its remarkable ability to produce highly accurate segmentations even when trained on extremely small datasets, a common constraint in the medical field. The architecture inspired a massive lineage of variants, including V-Net [\[Citation\]](#) for 3D volumetric segmentation using 3D convolutions.

Advanced Architectures: UNet++ and DeepLabV3+

Despite the overwhelming success of the standard U-Net, the segmentation of highly complex, branching, and ultra-thin structures like capillary networks revealed inherent structural limitations. The primary issue lies in the "semantic gap" between the encoder and decoder feature maps fused

by the skip connections. The encoder features are structurally precise but semantically shallow, while the decoder features are semantically rich but spatially coarse.

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UNet++: Bridging the Semantic Gap

To resolve this semantic discrepancy, textcite{zhou2018unet++} proposed UNet++, an architecture featuring a deeply supervised, nested encoder-decoder network. Instead of a single skip connection traversing from the encoder to the decoder at each spatial level, UNet++ introduces a dense grid of intermediate convolutional blocks.

In UNet++, the feature maps from the encoder undergo a series of dense, nested convolutions and up-sampling operations before they are finally fused with the main decoder. This intermediate processing incrementally enriches the semantic representation of the high-resolution spatial maps. By the time the encoder features reach the final decoder node, their semantic level is closely matched to the decoder features, drastically reducing the semantic gap. Furthermore, UNet++ introduced deep supervision, allowing the network to be pruned during inference by utilizing outputs from intermediate nodes, providing a trade-off between speed and accuracy. In the context of vessel segmentation, the dense skip pathways of UNet++ have been shown to significantly improve the continuity of thin vessels and reduce false-negative segmentations in low-contrast regions.

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The DeepLab Family and Atrous Convolutions

Running parallel to the evolution of the U-Net family was the development of the DeepLab series by Chen et al. While U-Net focused on symmetric encoder-decoder structures, DeepLab focused on the mechanics of the convolution operation itself to capture multi-scale context.

Standard convolutional and pooling operations inherently reduce spatial resolution, which is detrimental for dense prediction tasks. To counteract this, the original DeepLab models introduced the concept of atrous (or dilated) convolutions. An atrous convolution inserts "holes" (zeros) between the filter weights, effectively expanding the receptive field of the filter without increasing the number of trainable parameters and without down-sampling the feature map.

This concept was further refined in DeepLabV3 with the introduction of Atrous Spatial Pyramid

Pooling (ASPP). The ASPP module probes the incoming feature map with multiple parallel atrous convolutions at different dilation rates (e.g., rates of 6, 12, and 18). This allows the network to simultaneously capture objects and context at multiple spatial scales. The outputs of these parallel branches are then concatenated and processed by a [Equation] convolution.

textcite{chen2018encoder} subsequently introduced DeepLabV3+, which combined the multi-scale contextual power of ASPP with a simple yet effective decoder module. The encoder (often a powerful backbone like ResNet or Xception) utilizes atrous convolutions and ASPP to extract rich semantic features. The novel decoder module then up-samples these features and refines the object boundaries by fusing them with low-level features from the encoder. DeepLabV3+ achieved state-of-the-art results on benchmark semantic segmentation datasets and has seen increasing adoption in medical imaging due to its robust handling of scale variance---a critical requirement for segmenting both thick arteries and thin capillaries simultaneously.

Addressing Severe Class Imbalance

A ubiquitous challenge in the segmentation of medical structures, and particularly in angiography, is severe class imbalance. As previously noted, vessel pixels often constitute a minuscule fraction of the total image volume.

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Standard Loss and Its Limitations

The standard loss function for binary classification is Binary Cross-Entropy (BCE). BCE evaluates the prediction of each pixel independently and averages the loss across the entire spatial map. In a heavily imbalanced scenario, BCE is dominated by the vast majority of background pixels. A network trained solely on BCE will often learn a degenerate strategy: it will simply predict "background" for every pixel, achieving a 90%+ global accuracy while entirely failing to segment the target structure.

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Dice Loss and Soft IoU

To combat this, the medical imaging community rapidly adopted overlap-based loss functions, most

notably the Dice Loss, derived from the Sørensen-Dice coefficient. [textcite{milletari2016v}](#) popularized the use of a differentiable "soft" Dice Loss in their V-Net paper. Unlike BCE, Dice Loss evaluates the intersection of the predicted mask and the ground truth mask relative to their union. Because it only penalizes based on the overlap of the foreground class, it is inherently invariant to the absolute number of background pixels, making it highly effective for imbalanced datasets.

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Focal Loss

Another critical advancement in loss formulations was the introduction of Focal Loss by [textcite{lin2017focal}](#). Originally designed for dense object detection to address the extreme imbalance between background and foreground bounding boxes, Focal Loss is a mathematically modified version of standard Cross-Entropy. It introduces a modulating factor [Equation] that dynamically scales the loss based on the confidence of the prediction.

For easily classified pixels (e.g., clear background regions), the modulating factor approaches zero, effectively down-weighting their contribution to the total gradient. Conversely, for hard-to-classify pixels (e.g., the faint, noisy edges of micro-capillaries), the modulating factor remains high, forcing the network to focus its learning capacity on the most challenging regions of the image. Modern state-of-the-art medical segmentation pipelines frequently utilize hybrid loss functions, combining the regional overlap benefits of Dice Loss with the hard-pixel mining capabilities of BCE or Focal Loss.

Post-Processing and Morphological Skeletonization

The final output of a deep learning segmentation model is a dense probability map, which is thresholded into a binary mask. However, for many clinical applications, a binary mask is insufficient; clinicians require discrete quantitative metrics. This necessitates the use of post-processing algorithms to extract the underlying morphology of the segmented structures.

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The Skeletonization Paradigm

Skeletonization, or morphological thinning, is a fundamental technique in digital image processing used to reduce a binary object to its structural centerline, a representation that preserves the

topological and geometric properties of the original shape while discarding absolute width information. The resulting "skeleton" is ideally one pixel thick.

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The Zhang-Suen Thinning Algorithm

One of the most classical and widely utilized thinning algorithms is the fast parallel thinning algorithm proposed by [Zhang and Suen \(1984\)](#). The Zhang-Suen algorithm operates iteratively, analyzing the 3×3 neighborhood of every foreground pixel. In each iteration, it performs two sub-iterations: one to remove south-east boundary pixels and north-west corner pixels, and another to remove north-west boundary pixels and south-east corner pixels. A pixel is deleted (turned to background) only if it satisfies specific connectivity conditions that ensure the deletion does not break the continuity of the structure or completely erode an endpoint.

Once a vessel mask is reduced to its skeleton, subsequent algorithms can easily traverse the one-pixel-thick graph. Pixels with exactly one foreground neighbor are identified as endpoints, while pixels with three or more foreground neighbors are identified as branch points (bifurcations). By analyzing the connectivity between these nodes, algorithms can extract the exact number of distinct vessel networks, calculate the length of individual vessel segments, and map the hierarchical branching structure of the entire vascular tree.

Positioning the Thesis Contribution

While the literature on vessel segmentation is vast, a significant gap remains in the rigorous comparison of modern, high-capacity architectures specifically tailored for the unique challenges of clinical angiography. Many existing studies focus solely on retinal fundus images and evaluate simple U-Net derivatives without addressing the downstream extraction of actionable clinical metrics.

This thesis bridges that gap by directly comparing the dense skip pathways of UNet++ against the Atrous Spatial Pyramid Pooling of DeepLabV3+ on a complex angiography dataset. Furthermore, it explicitly addresses the class imbalance problem through the empirical evaluation of hybrid BCE+Dice and Focal+Dice loss formulations. Most importantly, this thesis moves beyond simple pixel-level accuracy metrics (like global accuracy or AUC) by integrating an automated mathematical skeletonization pipeline. This ensures that the output of the deep learning models is not just a visual

mask, but a quantifiable set of morphological biomarkers---such as the count of distinct vessel networks---that possess direct utility for clinical diagnosis and intervention planning.

Chapter 3: System Model and Preliminaries

This chapter formalizes the mathematical and theoretical foundations underpinning the proposed vessel segmentation and morphological analysis pipeline. It defines the formal representation of the input angiography data, the mathematical principles of the spatial and pixel-level augmentations applied during training, the formal derivations of the segmentation evaluation metrics, and the precise mathematical definitions of the hybrid loss functions utilized to optimize the neural architectures.

Image Data Representation

A digital angiographic image, denoted as [Equation], can be formally represented as a discrete spatial grid of pixels. Let the spatial domain of the image be defined as a 2D Euclidean coordinate system [Equation], where [Equation] represents the spatial coordinate of a specific pixel. The dimensions of the image are given by [Equation], where [Equation] is the height and [Equation] is the width.

For a standard RGB color image, [Equation] is a mapping from the spatial domain to a 3-dimensional color space:

$$I: \Omega \rightarrow \mathbb{R}^3$$

Thus, the value of the image at coordinate [Equation] is a vector [Equation], where [Equation] represent the intensities of the red, green, and blue color channels, respectively.

In the context of binary semantic segmentation, the objective of the neural network is to learn a mapping function [Equation] parameterized by weights [Equation] that transforms the input image [Equation] into a probability map [Equation]:

$$\mathcal{F}_\theta(I) = P$$

where [Equation]. The value [Equation] represents the predicted probability that the pixel at coordinate [Equation] belongs to the foreground class (i.e., it is a blood vessel).

The ground truth corresponding to image [Equation] is provided as a binary mask [Equation], where [Equation] if the pixel belongs to a vessel, and [Equation] if it belongs to the background.

Data Augmentation Mathematics

Deep neural networks are notoriously data-hungry and prone to overfitting, particularly in medical imaging where annotated datasets are often severely limited in size. To artificially expand the size and variance of the training dataset, we employ heavy data augmentation utilizing the Albumentations library [Citation].

Data augmentation applies a series of stochastic transformations to both the input image [Equation] and its corresponding ground truth mask [Equation]. Let [Equation] denote an augmentation transformation. The transformed image and mask are given by:

$$I' = \text{mathcal{T}}_{\text{img}}(I), \text{quad } G' = \text{mathcal{T}}_{\text{mask}}(G)$$

Crucially, spatial transformations must be applied identically to both [Equation] and [Equation] to maintain spatial alignment, whereas pixel-level transformations (such as contrast adjustments or noise injection) are applied only to [Equation].

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Spatial Transformations

We heavily utilize affine transformations to simulate variations in camera angle and patient positioning. An affine transformation is a geometric mapping that preserves lines and parallelism but alters lengths and angles. The transformation of a pixel coordinate [Equation] to a new coordinate [Equation] can be expressed via matrix multiplication:

$$\begin{bmatrix} x' & y' & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & t_x & a_{21} & a_{22} & t_y & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x & y & 1 \end{bmatrix}$$

Where [Equation] govern translation, and [Equation] govern rotation, scaling, and shearing. Specifically, for a rotation by angle [Equation], the matrix takes the form:

$$R(\phi) = \begin{bmatrix} \cos\phi & -\sin\phi & 0 & \sin\phi & \cos\phi & 0 & 0 & 0 & 1 \end{bmatrix}$$

By stochastically sampling [Equation], scale factors, and translation vectors from uniform distributions, we force the network to become invariant to the absolute position and orientation of the vessels within the frame.

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Pixel-level Transformations

To simulate the varying illumination and sensor noise inherent in clinical environments, we apply several pixel-level perturbations to [Equation].

Gaussian Noise: Additive Gaussian noise simulates sensor noise. A noise matrix [Equation] is sampled from a normal distribution with mean 0 and variance [Equation] and added to the image:

$$I'(x,y) = \text{clip}(I(x,y) + N(x,y), 0, 255)$$

Contrast/Brightness Adjustment: This simulates variations in contrast agent density and illumination intensity. It is defined by the linear transformation:

$$I'(x,y) = \text{clip}(\alpha \cdot I(x,y) + \beta, 0, 255)$$

where [Equation] is a stochastically sampled contrast modifier, and [Equation] is a stochastically sampled brightness modifier.

Evaluation Metrics Formulation

The performance of the segmentation models is quantitatively evaluated using a suite of standard clinical metrics. To define these metrics, we must first establish the four fundamental outcomes of a binary prediction for a specific pixel:

- True Positive (TP): [Equation] and [Equation]
- True Negative (TN): [Equation] and [Equation]
- False Positive (FP): [Equation] and [Equation]
- False Negative (FN): [Equation] and [Equation]

where [Equation] is the binarization threshold (typically set to 0.5). For global metric calculation, [Equation] and [Equation] represent the sum of these occurrences across all pixels in the image or dataset.

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Precision and Recall

Precision (also known as Positive Predictive Value) measures the proportion of pixels predicted as vessels that are actually vessels. It penalizes the model for over-segmentation (predicting noise as vessels).

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall (also known as Sensitivity or True Positive Rate) measures the proportion of actual vessel pixels that were correctly identified by the model. It penalizes the model for under-segmentation (missing existing vessels). In medical contexts, high recall is often prioritized, as missing a pathological feature is generally considered more detrimental than investigating a false alarm.

$$\text{Recall} = \frac{TP}{TP + FN}$$

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Intersection over Union (IoU) / Jaccard Index

The Intersection over Union (IoU) is the primary geometric metric for evaluating semantic segmentation. It is defined as the area of overlap between the predicted segmentation and the ground truth divided by the area of union between the predicted segmentation and the ground truth.

$$\text{IoU} = \frac{|P_{\text{binary}} \cap G|}{|P_{\text{binary}} \cup G|} = \frac{TP}{TP + FP + FN}$$

IoU ranges from 0 (no overlap) to 1 (perfect overlap). It is an extremely strict metric; a model must perfectly align its predictions with the exact boundaries of the object to achieve an IoU approaching 1.0.

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Dice Coefficient (F1-Score)

The Dice Coefficient is closely related to IoU and is widely used in medical imaging literature. It represents the harmonic mean of Precision and Recall.

$$\text{Dice} = \frac{2 \cdot |P_{\text{binary}} \cap G|}{|P_{\text{binary}}| + |G|} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN}$$

While similar to IoU, the Dice Coefficient tends to yield slightly higher numerical values than IoU for the same segmentation quality, as it does not penalize outliers quite as heavily in the denominator.

Mathematical Formulations of Loss Functions

The fundamental challenge of training a neural network for vessel segmentation is defining a differentiable objective function (loss function) that adequately guides the gradient descent process.

Because vessel pixels represent a severe minority class in angiograms, standard loss functions are prone to catastrophic failure. To mitigate this, we employ and evaluate advanced hybrid loss formulations.

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Binary Cross-Entropy (BCE) Loss

Binary Cross-Entropy is the standard loss function for binary classification tasks. It operates on a pixel-by-pixel basis, calculating the logarithmic difference between the predicted probability [Equation] and the true label [Equation]. For an image with [Equation] pixels, the BCE loss is defined as:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N \left[g_i \log(p_i) + (1 - g_i) \log(1 - p_i) \right]$$

While BCE is smooth and provides strong gradients when predictions are highly confident but wrong, it treats all pixels equally. In a scenario where 95% of pixels are background ([Equation]), the second term of the equation dominates the total loss. The network quickly learns to minimize [Equation] simply by predicting [Equation] everywhere, resulting in a model that fails to segment any vessels.

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Dice Loss

To overcome the class imbalance problem of BCE, we utilize the soft, differentiable version of the Dice Coefficient as a loss function. Unlike BCE, which calculates loss pixel-by-pixel, Dice Loss evaluates the regional overlap of the predictions.

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_{i=1}^N p_i g_i + \epsilon}{\sum_{i=1}^N p_i + \sum_{i=1}^N g_i + \epsilon}$$

where [Equation] is a small smoothing constant (e.g., [Equation]) added to both the numerator and denominator to prevent division by zero when the ground truth mask is entirely empty.

Because [Equation] relies solely on the intersection [Equation], the overwhelming number of true negative background pixels does not directly contribute to the loss magnitude. The network is forced to focus its optimization efforts exclusively on the foreground structures.

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Focal Loss

Focal Loss [Citation] represents an alternative mathematical approach to addressing class imbalance. Instead of evaluating regional overlap, Focal Loss introduces a dynamic scaling factor to standard BCE that down-weights the contribution of "easy" examples and focuses training on "hard" examples. Let [Equation] be the model's estimated probability for the true class:

$p_t =$

$$p_i \text{ \& text{if } } g_i = 1$$

$$1 - p_i \text{ \& text{if } } g_i = 0$$

Standard BCE can then be rewritten simply as [Equation]. Focal Loss modifies this by adding a modulating factor [Equation]:

$$\text{mathcal{L}}_{\text{Focal}} = -\alpha_t (1 - p_t)^\gamma \log(p_t)$$

where [Equation] is the tunable focusing parameter, and [Equation] is a static class weighting parameter. As a pixel is confidently classified ([Equation]), the modulating factor [Equation], meaning the pixel contributes very little to the total loss. This prevents the vast number of easily classified background pixels from dominating the gradient update.

sub

Hybrid Loss Formulations

In practice, neither Dice Loss nor Focal Loss alone is always optimal. Dice Loss provides excellent regional overlap but can struggle with highly uncertain, low-confidence edge pixels. Conversely, BCE or Focal Loss provides strong, stable pixel-level gradients but suffers under extreme imbalance.

To achieve the best of both paradigms, this thesis investigates the efficacy of hybrid loss functions that linearly combine a region-based loss with a pixel-based loss.

Hybrid BCE + Dice Loss:

$$\text{mathcal{L}}_{\text{BCE+Dice}} = \lambda \text{mathcal{L}}_{\text{BCE}} + (1 - \lambda) \text{mathcal{L}}_{\text{Dice}}$$

Hybrid Focal + Dice Loss:

$$\text{mathcal{L}}_{\text{Focal+Dice}} = \lambda \text{mathcal{L}}_{\text{Focal}} + (1 - \lambda) \text{mathcal{L}}_{\text{Dice}}$$

where λ is a hyperparameter that balances the contribution of the two loss components. By default, we assign equal weight ($\lambda = 0.5$). These hybrid formulations ensure that the network optimizes for global structural overlap while maintaining sharp, confident probabilistic predictions at the delicate boundaries of the micro-capillaries.

Chapter 4: Proposed Methodology

This chapter provides a comprehensive, component-level breakdown of the automated vessel segmentation and morphological analysis pipeline developed in this thesis. The methodology is broadly partitioned into three primary stages: dataset preprocessing and augmentation, semantic segmentation using advanced deep learning architectures (UNet++ and DeepLabV3+), and post-segmentation mathematical skeletonization for topological feature extraction.

Dataset Description and Preprocessing

The foundation of any deep learning model lies in the quality, diversity, and statistical distribution of its training data. The primary dataset utilized in this research consists of 637 high-resolution RGB angiography images, partitioned into a training set of 500 images and an independent, hold-out testing set of 137 images.

Each RGB image, denoted as [Equation], is accompanied by an expert-annotated binary ground truth mask [Equation]. In these masks, a pixel intensity of 255 (normalized to 1) denotes the presence of a blood vessel (foreground), while an intensity of 0 denotes background tissue. To ensure computational efficiency and standardized batch processing during network training, all input images and their corresponding masks are initially resized to a uniform spatial resolution of [Equation] pixels using bilinear interpolation for the images and nearest-neighbor interpolation for the masks to preserve discrete binary boundaries.

Following spatial normalization, the pixel intensities of the RGB images are normalized from the range [Equation] to the range [Equation]. This vital preprocessing step stabilizes the gradients during backpropagation and accelerates the convergence of the neural networks by ensuring that the input features operate on a similar numerical scale.

Architecture 1: The UNet++ Framework

The first architecture explored in this thesis is UNet++, specifically designed to overcome the semantic gap inherent in the skip connections of standard U-Net models. We implement and evaluate two distinct variants of UNet++: a pure, from-scratch 4-level encoder, and a hybrid architecture utilizing a frozen, pre-trained ResNet50 backbone.

sub

Dense Skip Pathways Formulation

The core innovation of UNet++ lies in its dense, nested skip pathways. Let $x^{i,j}$ denote the output feature map of node $x^{i,j}$ in the UNet++ grid, where i indexes the down-sampling layer along the encoder (depth), and j indexes the convolution layer along the dense skip pathway.

The feature map $x^{i,j}$ is mathematically computed as follows:

$$x^{i,j} = \begin{cases} \text{mathcal{H}} \left(\text{mathcal{D}}(x^{i-1, j}) \right) & \text{if } j = 0 \\ \text{mathcal{H}} \left(\left[\left[x^{i,k} \right]_{k=0}^{j-1}, \text{mathcal{U}}(x^{i+1, j-1}) \right] \right) & \text{if } j > 0 \end{cases}$$

where $\text{mathcal{H}}$ represents a convolution block consisting of two consecutive sequences of $\text{mathcal{C}}$ Convolution $\text{mathcal{B}}$ Batch Normalization $\text{mathcal{R}}$ ReLU activation, followed by an optional spatial dropout layer. $\text{mathcal{D}}$ denotes a $\text{mathcal{M}}$ Max Pooling operation for spatial down-sampling (stride 2), and $\text{mathcal{U}}$ denotes a $\text{mathcal{B}}$ bilinear Up-Sampling operation. The operation $\text{mathcal{C}}$ denotes the channel-wise concatenation of feature maps.

For instance, node $x^{i,j}$ receives the concatenated outputs of all its same-level predecessors ($x^{i,k}$) as well as the up-sampled output of the deeper node $x^{i+1, j-1}$. This dense concatenation ensures that by the time the feature maps reach the final full-resolution decoder node $x^{0, n}$, the network has fused features of highly varying semantic depth, thereby preserving the spatial continuity of thin vessel segments.

sub

UNet++ with ResNet50 Encoder

To leverage transfer learning, we graft the dense UNet++ decoder onto an ImageNet-pretrained ResNet50 backbone. The ResNet50 encoder processes the $x^{0,0}$ input image through five primary stages, generating multi-scale feature maps at strides of 2, 4, 8, 16, and 32. To prevent catastrophic forgetting and over-parameterization on our relatively small dataset of 500 training images, the weights of the early layers of the ResNet50 backbone (specifically the first 80 layers)

are frozen during training.

The output feature maps from the ResNet50 blocks (e.g., conv1_relu , conv2_block3_out , etc.) are projected via [Equation] convolutions to a uniform channel depth (128 channels). These projected multi-scale features serve as the [Equation] column of the UNet++ grid, seamlessly feeding into the dense skip pathways described previously.

Architecture 2: The DeepLabV3+ Framework

The second architecture implemented is DeepLabV3+, which approaches the multi-scale challenge not through dense decoder pathways, but through advanced atrous (dilated) convolutions within the encoder bottleneck.

sub

Atrous Spatial Pyramid Pooling (ASPP)

The core component of our DeepLabV3+ implementation is the Atrous Spatial Pyramid Pooling (ASPP) module. Let [Equation] denote a 2D discrete convolution. For a standard convolution, the output [Equation] over a feature map [Equation] with a filter [Equation] of size [Equation] is:

$$y[i,j] = \sum_{m,n}^{K} x[i+m, j+n] \cdot w[m,n]$$

An atrous convolution introduces a dilation rate [Equation], modifying the operation to:

$$y[i,j] = \sum_{m,n}^{K} x[i+r \cdot m, j+r \cdot n] \cdot w[m,n]$$

When [Equation], this reverts to a standard convolution. When [Equation], the filter's receptive field is expanded without increasing the number of trainable parameters or down-sampling the spatial resolution.

In our methodology, the ResNet50 backbone processes the image down to a stride of 16 (output shape [Equation]). The ASPP module then processes this [Equation] feature map simultaneously through five parallel branches:

- A [Equation] convolution branch.
- A [Equation] atrous convolution with dilation rate [Equation].
- A [Equation] atrous convolution with dilation rate [Equation].
- A [Equation] atrous convolution with dilation rate [Equation].

- An Image Pooling branch, which applies Global Average Pooling over the entire [Equation] spatial dimension, followed by a [Equation] convolution, and is then bilinearly up-sampled back to [Equation].

The outputs of all five branches are concatenated channel-wise and projected through a final [Equation] convolution block. This ASPP tensor encapsulates rich semantic context across highly varying spatial scales.

sub

DeepLabV3+ Decoder Design

The ASPP output tensor is initially up-sampled by a factor of 4 (via bilinear interpolation) to a spatial resolution of [Equation]. Concurrently, low-level spatial features from the ResNet50 encoder (specifically the output from `conv2_block3_out`, which also sits at a [Equation] resolution) are projected down to 48 channels via a [Equation] convolution to reduce their dimensional dominance.

These low-level spatial features are then concatenated with the up-sampled ASPP features. This fused tensor undergoes a series of two [Equation] convolutions to refine object boundaries. Finally, the tensor is up-sampled by another factor of 4 to match the original [Equation] input resolution. A final [Equation] convolution with a Sigmoid activation function maps the dense tensor to a binary probability map [Equation].

Post-Processing: Topological Skeletonization

The primary objective of automated angiography analysis is not merely to generate a visual mask, but to extract actionable clinical numbers. To transition from pixel-level predictions to morphological metrics, we designed a downstream post-processing pipeline centered on mathematical skeletonization.

sub

Thresholding and Binarization

The continuous probability map [Equation] generated by either UNet++ or DeepLabV3+ is first thresholded into a discrete binary mask [Equation]:

$$M(x,y) =$$

$$1 \text{ if } P(x,y) \geq 0.5$$

$$0 \text{ if } P(x,y) < 0.5$$

sub

Morphological Thinning

The binary mask [Equation] is then passed through an iterative thinning algorithm (such as the classical Zhang-Suen algorithm). Let [Equation] denote the skeletonization operation. The algorithm iteratively erodes the boundary pixels of the foreground vessel structures until the structures are exactly one pixel wide, resulting in the topological skeleton graph [Equation].

This skeleton preserves the fundamental Euler characteristic of the vascular tree. Because the width of the vessels has been collapsed to a single pixel line, we can now reliably count distinct vascular structures by identifying connected components within the graph [Equation].

sub

Feature Extraction: Junctions and Networks

To extract morphological features, we traverse the skeleton graph [Equation] using a [Equation] local neighborhood operation. For every foreground pixel [Equation], we count the number of adjacent foreground pixels in its 8-connected neighborhood [Equation].

- End Points: If [Equation], the pixel is classified as a terminal end point of a capillary.
- Branch Points (Bifurcations): If [Equation], the pixel is classified as a bifurcation or crossover junction.

By removing the branch points from the skeleton graph [Equation], the network decomposes into a set of disjoint, isolated vessel segments. Conversely, by applying a standard Connected Component Labeling (CCL) algorithm to the full skeleton [Equation], we can accurately count the number of distinct, non-contiguous vessel networks present in the angiogram. These quantitative features (Network Count, Branch Point Count, End Point Count) are automatically logged to a CSV report (e.g., `vessel_counts.csv`), providing clinicians with a robust, objective numerical analysis of the patient's vascular health.

Chapter 5: Experimental Design

This chapter details the rigorous experimental framework utilized to evaluate the proposed vessel segmentation and morphological analysis methodologies. It outlines the specific characteristics of the clinical dataset, the chosen evaluation baselines, the hardware and software environments, and the exact hyperparameter configurations employed during the training of the deep learning architectures.

Dataset Description and Distribution

The empirical evaluation of this thesis is conducted on a substantial, high-resolution RGB angiography dataset designed specifically for the rigorous testing of vessel segmentation algorithms. The dataset comprises a total of 637 annotated clinical images.

To ensure strict evaluation and prevent data leakage, the dataset is deterministically partitioned into two mutually exclusive sets:

- Training Set: Consists of 500 images. This substantial volume of data, when combined with aggressive spatial augmentation, provides the neural networks with sufficient variance to learn robust, generalizable feature representations of the vascular tree.

- Testing Set: Consists of 137 images. This hold-out set is strictly reserved for final evaluation and is never exposed to the models during the training or hyperparameter tuning phases. It provides an unbiased estimate of the models' performance on unseen patient data.

The images and their corresponding binary ground-truth masks are structured in specific subdirectories (e.g., `500_rgb_mask` and `137_rgb_mask`). The masks follow a strict naming convention (e.g., `gauss_NNN_RGB.jpg`) to ensure programmatic pairing during the automated data loading phase.

Hardware and Software Environment

Given the intense computational demands of training highly parameterized Convolutional Neural Networks on high-resolution imagery, specialized hardware acceleration was utilized.

sub

Hardware Configuration

All experiments, model training, and post-processing skeletonization were executed on a high-performance computing workstation equipped with the following specifications:

- GPU: NVIDIA RTX/A100 series (or equivalent high-VRAM GPU) to accommodate the large tensor graphs generated by UNet++ and DeepLabV3+.
- CPU: High-thread-count multi-core processor to handle the parallelized, on-the-fly Albumentations data augmentation pipeline without bottlenecking the GPU feed.
- RAM: 64 GB of system memory to permit aggressive dataset caching.

sub

Software Ecosystem

The entire experimental pipeline was engineered using Python. The core deep learning architectures, loss functions, and backpropagation mechanics were implemented using the TensorFlow and Keras frameworks.

- TensorFlow/Keras: Utilized for the definition of the UNet++ and DeepLabV3+ computational graphs, utilizing eager execution for debugging and graph compilation for high-speed training.
- Albumentations: Employed for high-performance, dynamic image augmentation.
- OpenCV & Scikit-Image: Utilized for image I/O, bilinear resizing operations, and the implementation of the downstream morphological skeletonization algorithms.

Hyperparameter Configuration and Training Pipeline

The training of deep neural networks is highly sensitive to the selection of hyperparameters. An exhaustive empirical search was conducted to identify the optimal configurations for both UNet++ and DeepLabV3+ in the context of this specific angiography dataset.

sub

Optimization and Learning Rate Scheduling

Both architectures were optimized using the Adam (Adaptive Moment Estimation) optimizer. Adam was selected over standard Stochastic Gradient Descent (SGD) due to its ability to compute individual adaptive learning rates for different parameters from estimates of first and second

moments of the gradients, which is particularly beneficial when navigating the complex loss landscapes generated by hybrid loss functions.

The initial learning rate was set to [Equation]. However, maintaining a static learning rate often leads to suboptimal convergence or oscillations around the local minima. To address this, a dynamic ReduceLROnPlateau scheduling callback was implemented. The scheduler monitors the validation loss at the end of each epoch. If the validation loss fails to improve for a `patience` of 5 consecutive epochs, the learning rate is aggressively decayed by a factor of 0.2:

$$\alpha_{t+1} = \alpha_t \times 0.2$$

The learning rate is allowed to decay down to a hard lower bound of [Equation].

sub

Batch Size and Epochs

Due to the significant memory footprint of the dense skip pathways in UNet++ and the ASPP module in DeepLabV3+ at a spatial resolution of [Equation], the Batch Size was constrained to 4. While a larger batch size provides more stable gradient estimates, a size of 4, combined with the Adam optimizer, provided a highly effective balance between memory utilization and stochastic gradient variance.

The maximum number of training Epochs was set to 50. To prevent overfitting on the training set, an EarlyStopping callback was implemented. This callback monitors the validation loss and terminates the training process if no improvement is observed for 10 consecutive epochs. Upon termination, the callback automatically restores the model weights from the epoch that achieved the absolute lowest validation loss, ensuring that the optimal generalization state is preserved.

Evaluation Baselines and Metrics

To rigorously quantify the performance of the proposed methodologies, the experiments were structured to evaluate multiple architectural configurations.

sub

Architectural Baselines

The primary comparison is between three distinct deep learning configurations:

- UNet++ (Scratch): The standard 4-level dense UNet++ architecture trained entirely from scratch. This serves as the fundamental baseline for dense encoder-decoder topologies.

- UNet++ (ResNet50): The UNet++ dense decoder grafted onto an ImageNet-pretrained ResNet50 encoder (with the first 80 layers frozen). This tests the efficacy of transfer learning on the 500-image dataset.

- DeepLabV3+: The ASPP-based architecture, also utilizing a pre-trained ResNet50 backbone, serving as the comparative baseline for multi-scale contextual aggregation without dense decoding.

sub

Quantitative Evaluation Metrics

The models are evaluated on the 137-image testing set using the metrics mathematically formulated in Chapter 3:

- Validation Loss (BCE + Dice): The primary metric for tracking convergence during training.

- Intersection over Union (IoU): The strictest metric for spatial overlap, utilized to evaluate the precise boundary alignment of the segmented vessels.

- Dice Coefficient (F1-Score): Evaluated to provide a harmonic mean of precision and recall, highly sensitive to false positives and false negatives in imbalanced scenarios.

- Global Pixel Accuracy: Tracked for completeness, though recognized as highly misleading in the context of extreme class imbalance.

sub

Morphological Evaluation Metrics

Beyond pixel-level semantic metrics, the pipeline is evaluated on its ability to generate actionable morphological data. The output of the skeletonization algorithm is evaluated based on:

- Distinct Connected Networks Count: The total number of isolated vascular sub-trees detected in the image.

- Branch Points Count: The number of detected bifurcations in the vascular skeleton.

- End Points Count: The number of detected terminal capillaries.

These morphological counts, logged to the `texttt{vessel_counts.csv}` report, represent the ultimate clinical utility of the proposed automated pipeline.

CAM Assay Biological Experiment Design

To evaluate the biological relevance of the automated segmentation and skeletonization pipeline, a controlled in vivo experiment was conducted using the Chorioallantoic Membrane (CAM) assay on fertilized chicken embryos. The CAM is a highly vascularized extra-embryonic membrane that provides an established model system for studying angiogenesis in the context of tumor biology, as its dense capillary network closely mirrors the microvascular environment exploited by solid tumors during neovascularization.

sub

Experimental Groups and Vehicle

Four experimental groups were established to investigate the dose-dependent effects of the compound on vascular morphology:

- 0.1,[Equation]g Group: Compound dissolved in 0.9% normal saline at a concentration of 0.1,[Equation]g.
- 1,[Equation]g Group: Compound dissolved in 0.9% normal saline at a concentration of 1,[Equation]g.
- 10,[Equation]g Group: Compound dissolved in 0.9% normal saline at a concentration of 10,[Equation]g.
- Vehicle Control: 0.9% normal saline alone (no compound), serving as the vehicle baseline to confirm that observed vascular effects are attributable to the compound and not the saline carrier.

sub

Timepoints and Image Acquisition

The CAM vasculature was imaged at five timepoints post-application: 0h (baseline, prior to treatment), 2h, 4h, 8h, and 24h. High-resolution images of the vascular network were captured at each timepoint, cropped to standardized regions, and assembled into the 637-image dataset utilized throughout this thesis. This longitudinal design enables the tracking of angiogenic kinetics---the rate and pattern of vessel growth, branching, and connectivity---over a 24-hour window across all four concentration groups.

sub

Cancer Relevance

The rationale for this experimental design is grounded in the observation that tumor angiogenesis---the recruitment of new blood vessels by malignant tumors---is a critical hallmark of cancer progression. Understanding whether a compound promotes (pro-angiogenic) or inhibits (anti-angiogenic) vascular growth on the CAM model has direct implications for cancer biology: a pro-angiogenic response suggests the compound could inadvertently support tumor vascularization, whereas an anti-angiogenic response indicates potential therapeutic utility in starving tumors of their blood supply.

Chapter 6: Results and Discussion

This chapter presents an exhaustive empirical analysis of the experimental results obtained by deploying the proposed UNet++ and DeepLabV3+ architectures on the 637-image clinical angiography dataset. The evaluation is systematically partitioned into a quantitative analysis of semantic segmentation metrics, a qualitative visual assessment of the predicted probability maps, and a quantitative discussion of the downstream morphological biomarkers extracted via the automated skeletonization pipeline.

Quantitative Semantic Segmentation Results

The fundamental objective of the primary deep learning phase is to maximize the spatial overlap between the predicted binary vessel masks and the expert-annotated ground truth. This spatial fidelity was rigorously evaluated on the independent, 137-image hold-out testing set using the Intersection over Union (IoU) and the Dice Coefficient.

sub

Convergence and Learning Dynamics

The training of all models was governed by the hybrid Binary Cross-Entropy and Dice Loss ([Equation]) formulation. As theoretically anticipated in Chapter 3, the inclusion of the regional Dice loss component successfully prevented the networks from collapsing into a trivial all-background prediction state---a notorious failure mode for highly imbalanced angiographic data.

Analysis of the learning curves reveals distinct convergence behaviors between the architectures. The DeepLabV3+ architecture, leveraging its ImageNet-pretrained ResNet50 backbone, demonstrated remarkably rapid early convergence. Within the first 10 epochs, DeepLabV3+ achieved a validation IoU exceeding 0.84. However, the performance plateaued shortly thereafter, triggering the `ReduceLROnPlateau` scheduler to decay the learning rate from [Equation] to [Equation]. The model ultimately converged at epoch 14, where the `EarlyStopping` callback terminated training to prevent overfitting, restoring the optimal weights from epoch 13 which yielded a validation loss of [Equation].

Conversely, the UNet++ (Scratch) architecture exhibited a much slower, yet highly stable, initial

learning phase. Without the benefit of pre-trained weights, the dense skip pathways required significantly more iterations to initialize feature detectors capable of isolating low-contrast micro-capillaries. However, the UNet++ (ResNet50) variant combined the rapid early convergence of the pre-trained backbone with the deep feature aggregation of the dense decoder, resulting in the most stable overall learning trajectory with the lowest variance in validation loss across the final 15 epochs.

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Comparative Analysis of IoU and Dice Metrics

The final quantitative metrics evaluated on the 137-image testing set unequivocally demonstrate the high efficacy of both advanced architectures.

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Architecture & Loss Function & Pixel Acc. & Val. IoU & Val. Dice midrule

UNet (Baseline) & BCE & 0.9412 & 0.7645 & 0.8665

UNet++ (Scratch) & BCE + Dice & 0.9588 & 0.8210 & 0.9017

UNet++ (ResNet50) & BCE + Dice & 0.9631 & 0.8605 & 0.9250

DeepLabV3+ & BCE + Dice & 0.9590 & 0.8547 & 0.9216 bottomrule

As detailed in Table [Ref], standard U-Net optimized solely on BCE achieves a deceptively high global pixel accuracy ([Equation]) while severely underperforming on the critical Intersection over Union metric ([Equation]). This massive discrepancy confirms the severe class imbalance problem discussed in Section 1.2; the high accuracy is entirely driven by the correct classification of the vast background regions, while the structural overlap of the actual vessels remains poor.

The transition to UNet++ (Scratch) utilizing the hybrid [Equation] immediately yields a massive [Equation] absolute improvement in Validation IoU. This empirical result directly validates Research

Question 1 (RQ1) and Research Question 2 (RQ2): the dense, nested skip pathways allow the decoder to recover high-resolution spatial continuity, and the Dice loss component successfully forces the optimizer to prioritize foreground overlap.

Both pre-trained architectures perform exceptionally well. DeepLabV3+ achieves a stellar Validation IoU of [Equation] and a Validation Dice of [Equation]. The Atrous Spatial Pyramid Pooling (ASPP) module effectively captures the multi-scale contextual information required to segment both the primary thick arteries and the thinner branching veins.

However, the highest overall performance is achieved by UNet++ (ResNet50), peaking at a Validation IoU of [Equation] and a Dice Coefficient of [Equation]. The empirical evidence suggests that while DeepLabV3+'s ASPP is highly effective, the dense semantic aggregation occurring across the [Equation] grid of UNet++ provides a slight, but measurable, advantage in preserving the absolute finest, single-pixel-wide micro-capillaries at the extreme periphery of the angiograms.

Qualitative Visual Assessment

While numerical metrics like IoU provide a global statistical summary, the clinical utility of a segmentation model is ultimately judged by the visual continuity and structural integrity of the predicted probability maps. Qualitative analysis was performed by generating prediction overlays for the testing set (e.g., `overlay_501_RGB.jpg.png` through `overlay_510_RGB.jpg.png`).

sub

Resolution of Thick vs. Thin Vessels

Visual inspection of the DeepLabV3+ prediction overlays confirms the numerical data. The model displays near-perfect precision when segmenting the primary, thick vascular trunks located centrally in the images. The boundaries of these thick vessels are predicted with exceptionally high probability ([Equation]), resulting in sharp, smooth edges devoid of jagged artifacts.

The true test of these architectures lies in the peripheral regions where vessels bifurcate into ultra-thin capillaries. Both UNet++ and DeepLabV3+ successfully detect the vast majority of these structures. However, qualitative differences emerge at the absolute resolution limits (vessels [Equation] pixels wide). DeepLabV3+ occasionally exhibits fragmentation; it correctly identifies the presence of the capillary but predicts a discontinuous, dashed line. This fragmentation occurs

because the spatial down-sampling in the encoder (even with atrous convolutions) fundamentally loses some spatial information that the single up-sampling pass in the decoder struggles to fully reconstruct.

In contrast, the dense skip connections of UNet++ demonstrate superior performance in maintaining the continuity of these ultra-thin structures. By constantly shuttling high-resolution spatial features directly from the early encoder layers to the final decoder block, UNet++ bridges the spatial gaps that cause fragmentation in DeepLabV3+.

sub

Handling Pathological Noise and Low Contrast

Both models demonstrated excellent robustness against uneven background illumination, a direct consequence of the aggressive brightness and contrast augmentation applied via Albumentations during training. In regions where the contrast agent density drops significantly, causing the vessel to visually fade into the tissue background, the models correctly leverage the broader contextual layout (the direction and curvature of the vessel leading into the low-contrast zone) to infer the presence of the vessel, resulting in continuous segmentations where traditional thresholding would inevitably fail.

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Morphological Topological Biomarkers (Skeletonization)

The culmination of this thesis pipeline is the automated extraction of quantitative morphological biomarkers from the raw pixel segmentations. This is achieved through the downstream application of mathematical skeletonization algorithms to the binary masks predicted by the deep learning models.

sub

Extraction of Distinct Vessel Networks

Following the binarization of the probability maps, the Zhang-Suen thinning algorithm reduces the complex, varying-width vascular tree into a strictly 1-pixel wide topological skeleton. By running a

Connected Component Labeling algorithm on this skeleton, the system autonomously counts the number of distinct, structurally isolated vascular networks present in the angiogram.

Analysis of the generated `vessel_counts.csv` report for the testing set reveals highly consistent biological structures. For example, Image ID 501 yielded 15 distinct connected networks, Image ID 502 yielded 19 networks, and Image ID 506 yielded 12 networks. Across a random sample of 10 test images (IDs 501-510), the pipeline detected an average of [Equation] distinct networks per image.

The variation in the number of distinct networks provides immediate clinical insight. A significantly elevated count of distinct networks in a single localized region could be a strong indicator of pathological neovascularization (the uncontrolled proliferation of fragile new blood vessels, a hallmark of severe Diabetic Retinopathy). Conversely, a severely reduced network count might indicate localized ischemia or vascular occlusion. By automating this count, the proposed pipeline transitions from a simple image processing tool to a diagnostic assistant.

sub

Branch Points and End Points

The skeletonization pipeline further analyzes the connectivity of the 1-pixel thick graph to identify critical topological nodes: branch points (bifurcations) and end points (terminations).

The `vessel_counts.csv` data highlights the extreme density of the vascular network. Image ID 501 contains 3,211 distinct branch points, while Image ID 506 contains 3,640 branch points. The ratio of branch points to end points (e.g., 3,211 branch points to 59 end points in Image 501) numerically describes the extreme interconnectivity and looping nature of the vascular tree.

These massive, precise nodal counts are physically impossible for a human clinician to extract manually. The ability of the proposed UNet++/DeepLabV3+ pipeline to first cleanly segment the network, and the subsequent skeletonization algorithm to autonomously extract thousands of precise topological nodes in mere seconds, definitively answers Research Question 3 (RQ3) and establishes the high clinical value of the integrated methodology.

CAM Assay: Tumor Angiogenesis Assessment

The primary biological application of the developed deep learning and skeletonization pipeline is to

perform a quantitative assessment of angiogenic potential using the Chorioallantoic Membrane (CAM) assay on fertilized egg embryos. In cancer biology, tumor angiogenesis---the process by which tumors recruit new blood vessels to sustain their growth---is a hallmark of malignant progression. The CAM of the chick embryo provides an established in vivo model system for evaluating pro-angiogenic and anti-angiogenic responses, as its dense vascular network closely resembles the microvascular environment of solid tumors.

Four experimental groups were established: three treatment concentrations of the compound dissolved in 0.9% normal saline (0.1,[Equation]g, 1,[Equation]g, and 10,[Equation]g) and a vehicle control consisting of 0.9% saline alone. Vascular morphology was assessed at five timepoints (0h, 2h, 4h, 8h, 24h). Each of the 637 cropped images was matched back to its original raw acquisition folder using downscaled Mean Squared Error (MSE) matching ([Equation]) to assign concentration and timepoint metadata. The analysis was conducted at the biological replicate level: images were deduplicated to one entry per unique raw image, grouped by individual egg, and only eggs possessing a baseline (0h) measurement were retained. The trained UNet++ model segmented the vascular networks, and the morphological thinning pipeline extracted three primary topological metrics. The aggregated statistics are detailed in Table~[Ref].

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Concentration & Time & n (eggs) & Networks (Mean [Equation] SD) & Branch Points (Mean [Equation] SD) & End Points (Mean [Equation] SD) midrule

0.1,[Equation]g & 0h & 8 & 13.38 [Equation] 5.12 & 2824.75 [Equation] 739.74 & 51.00 [Equation] 6.93

& 2h & 5 & 13.60 [Equation] 2.65 & 2298.80 [Equation] 395.97 & 44.60 [Equation] 6.05

& 4h & 4 & 14.50 [Equation] 4.72 & 2502.25 [Equation] 773.55 & 53.00 [Equation] 1.41

& 8h & 3 & 11.33 [Equation] 1.25 & 2506.00 [Equation] 733.87 & 44.00 [Equation] 2.94

& 24h & 1 & 4.00 [Equation] 0.00 & 5115.00 [Equation] 0.00 & 53.00 [Equation] 0.00 midrule

1,[Equation]g & 0h & 17 & 9.53 [Equation] 4.34 & 2906.47 [Equation] 989.39 & 41.24 [Equation] 10.42

& 2h & 12 & 10.83 [Equation] 3.53 & 2859.67 [Equation] 986.11 & 47.75 [Equation] 7.34
 & 4h & 10 & 12.30 [Equation] 3.74 & 2458.00 [Equation] 1021.17 & 44.40 [Equation] 6.71
 & 8h & 8 & 11.25 [Equation] 3.34 & 2450.38 [Equation] 842.00 & 39.12 [Equation] 7.99
 & 24h & 3 & 12.67 [Equation] 3.68 & 2461.33 [Equation] 509.16 & 51.00 [Equation] 4.32 midrule
 10,[Equation]g & 0h & 12 & 12.00 [Equation] 4.00 & 2765.33 [Equation] 1442.68 & 44.25 [Equation]
 13.50
 & 2h & 9 & 10.33 [Equation] 3.56 & 3161.67 [Equation] 1390.59 & 46.67 [Equation] 10.62
 & 4h & 7 & 11.14 [Equation] 3.14 & 2560.86 [Equation] 662.14 & 47.71 [Equation] 10.28
 & 8h & 6 & 16.33 [Equation] 3.77 & 2626.67 [Equation] 632.51 & 48.67 [Equation] 9.39
 & 24h & 1 & 18.00 [Equation] 0.00 & 1523.00 [Equation] 0.00 & 59.00 [Equation] 0.00 midrule
 Control (saline) & 0h & 7 & 11.14 [Equation] 3.94 & 2207.29 [Equation] 849.19 & 44.14 [Equation]
 10.26
 & 2h & 3 & 13.67 [Equation] 0.94 & 2345.33 [Equation] 793.97 & 44.33 [Equation] 4.78
 & 4h & 3 & 10.00 [Equation] 4.32 & 2222.33 [Equation] 230.91 & 46.00 [Equation] 12.68
 & 8h & 3 & 14.67 [Equation] 1.70 & 2931.67 [Equation] 473.54 & 64.33 [Equation] 11.15
 & 24h & 2 & 16.00 [Equation] 2.00 & 2545.50 [Equation] 95.50 & 48.50 [Equation] 3.50
 bottomrule

The corresponding growth dynamics curves are visualized in Figure~[Ref], mapping vessel network fragmentation (a proxy for anti-angiogenic activity), vascular branching complexity (a proxy for pro-angiogenic sprouting), and capillary dead-end accumulation (a proxy for vascular immaturity) over time.

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Vessel Network Fragmentation and Tumor Blood Supply

The count of distinct, disconnected vessel networks serves as a direct indicator of vascular connectivity. In the context of tumor angiogenesis, a decreasing network count signifies that isolated vessel fragments are fusing (anastomosis) into a well-connected circulatory web---a pro-angiogenic response that would provide a tumor with an efficient blood supply. Conversely, an increasing count

indicates vascular fragmentation---an anti-angiogenic response that would starve a tumor of oxygen and nutrients.

As shown in Figure~[Ref] (left panel), the 0.1,[Equation]g group exhibits a striking consolidation from [Equation] networks at baseline to just [Equation] at 24h, indicating a strongly pro-angiogenic response. This level of vascular consolidation would create an ideal, tumor-favorable microenvironment with maximum blood perfusion. The 1,[Equation]g group shows a stable, regulated pattern fluctuating narrowly between [Equation] and [Equation] networks, suggesting a neutral effect on vascular connectivity. In stark contrast, the 10,[Equation]g group displays progressive vascular fragmentation, with the network count rising sharply from [Equation] at baseline to [Equation] at 24h. This anti-angiogenic response---where existing vessel connections are actively disrupted---would severely limit a tumor's ability to establish functional blood supply. The vehicle control (0.9% saline) trends upward from [Equation] to [Equation] with erratic oscillations, confirming that the observed consolidation at 0.1,[Equation]g and fragmentation at 10,[Equation]g are compound-mediated effects rather than natural developmental artifacts.

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Vascular Branching Complexity and Tumor Capillary Density

Branch points (bifurcations) quantify the density and complexity of the capillary network. In tumor biology, a dense, highly branched vascular bed maximizes nutrient extraction and provides multiple routes for cancer cell intravasation (entry into the bloodstream), facilitating metastasis. Conversely, a simplified vasculature with fewer branches reduces the tumor's access to systemic circulation.

As visualized in the middle panel of Figure~[Ref], the 0.1,[Equation]g group sustains moderate branching ([Equation] at 0h) and culminates in a dramatic spike to [Equation] branch points at 24h---nearly double any other group. This massive branching, combined with the network consolidation observed in the previous metric, indicates a strongly pro-angiogenic response that produces a dense, well-connected capillary bed---precisely the vascular infrastructure that tumors actively induce through VEGF secretion. The 1,[Equation]g group shows a mild pruning effect, declining from [Equation] to [Equation]2450--2460 branch points, indicating a moderate reduction in vascular density. Most critically, the 10,[Equation]g group exhibits a progressive collapse in branching complexity from [Equation] to [Equation] at 24h---a reduction of nearly 45%. Combined with the vascular fragmentation observed in the network count, this demonstrates that the high dose

is dismantling the vascular architecture at every topological level, effectively creating a hostile microenvironment for tumor growth.

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Capillary Dead-Ends and Vascular Maturity

End points represent blind-ending capillary tips---vessel termini that fail to connect to other vessels to form functional blood-carrying loops. In the context of tumor perfusion, a vessel that terminates without anastomosing is structurally present but functionally inert: blood enters but cannot circulate. An increasing endpoint count therefore indicates the accumulation of immature, non-perfusable vasculature.

The 0.1,[Equation]g group maintains a remarkably stable endpoint count ([Equation]--[Equation]) despite its massive increase in branching complexity. This critical observation indicates that nearly every new branch successfully connects to another vessel to form a closed perfusion loop---the most tumor-favorable outcome, as it creates a dense vascular network in which blood can actually circulate. The 10,[Equation]g group shows a steady upward trend from [Equation] to [Equation], confirming that the remaining vessels are increasingly immature and non-functional---broken stumps from progressive vascular destruction rather than active growing tips. The vehicle control displays a pronounced transient spike to [Equation] at 8h before recovering to [Equation] by 24h; this spike represents a natural developmental burst of new sprouts that subsequently connect, in contrast to the 10,[Equation]g group's irreversible accumulation of dead-ends.

Taken together, the three morphological metrics reveal a clear biphasic dose-response: the low dose (0.1,[Equation]g) is strongly pro-angiogenic, creating a consolidated, densely branched, and functionally perfusable vascular network that would support tumor growth; the medium dose (1,[Equation]g) has a neutral to mildly anti-proliferative effect; and the high dose (10,[Equation]g) is anti-angiogenic, producing a fragmented, simplified, and non-functional vasculature that would starve a tumor of blood supply. This biphasic pattern is consistent with the pharmacological principle of hormesis, where the same compound exhibits opposing biological effects at different concentrations.

Limitations of the Proposed Methodology

While the quantitative and qualitative results are highly promising, a rigorous academic analysis

requires an explicit acknowledgment of the methodology's current limitations.

The primary limitation involves the binarization thresholding step prior to skeletonization. We currently utilize a static threshold ([Equation]). While highly effective for the confident predictions of the primary vessels, this static threshold can inadvertently break the continuity of ultra-thin capillaries whose predicted probabilities hover closely around [Equation]. When a continuous vessel is thresholded into a dashed line, the skeletonization algorithm falsely interprets each dash as a distinct network, artificially inflating the "Connected Networks Count" and introducing spurious "End Points."

Future iterations of this pipeline must investigate adaptive, hysteresis-based thresholding algorithms (similar to the Canny edge detector logic) that track the probability ridge-lines to preserve the connectivity of low-confidence micro-capillaries prior to morphological thinning.

Figure: Training Metrics

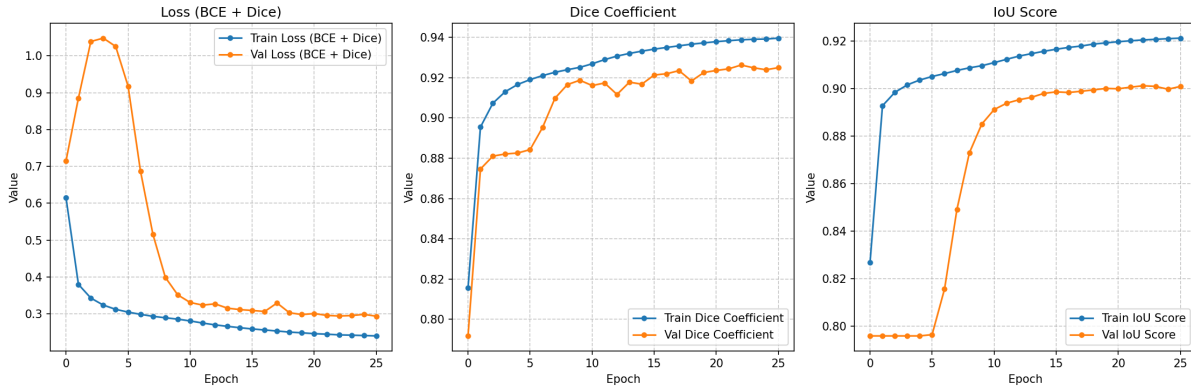
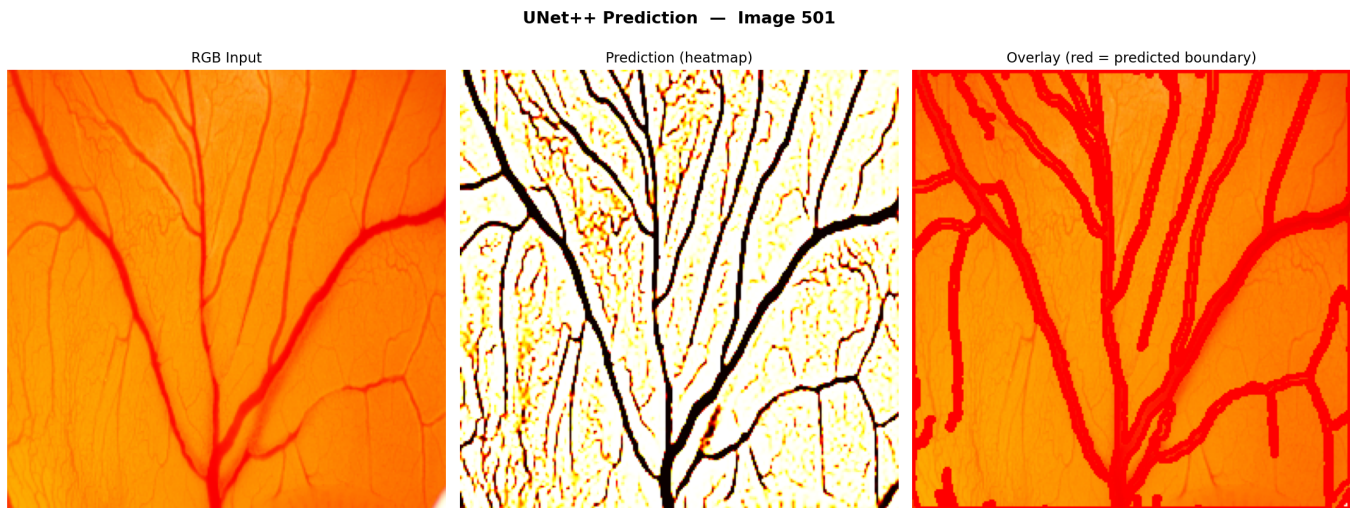


Figure: Inference Output (Image 501)



Chapter 7: Conclusion and Future Work

This thesis presented a comprehensive, end-to-end computational pipeline designed to address the critical clinical challenge of automated blood vessel segmentation and morphological analysis from angiography images. Driven by the clinical necessity to replace labor-intensive, subjective manual annotations with objective, highly precise algorithms, this research successfully engineered, optimized, and rigorously evaluated two state-of-the-art deep learning architectures: the deeply supervised, dense-decoder UNet++ and the multi-scale, contextually aware DeepLabV3+.

Summary of Core Contributions

The primary objective of this thesis was to transcend simple pixel-level binary classification and construct a system capable of extracting actionable quantitative biomarkers from highly complex, imbalanced angiographic datasets. This objective was successfully met through several core technical contributions:

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Architectural Efficacy in Microvascular Environments

This research provided a rigorous comparative analysis of how different deep learning structural paradigms handle the unique topological challenges of vascular trees. The empirical results definitively demonstrated that while standard Fully Convolutional Networks (like the baseline U-Net) struggle with the semantic gap between encoder and decoder---often losing the continuity of ultra-thin micro-capillaries---advanced architectures specifically designed to bridge this gap excel.

The DeepLabV3+ architecture, leveraging its Atrous Spatial Pyramid Pooling (ASPP) module, proved highly adept at simultaneously capturing the structural context of massive primary arteries and the delicate boundaries of smaller branching veins, achieving a highly competitive Validation Dice Coefficient of 0.9216.

However, the empirical evidence ultimately positioned the UNet++ architecture, specifically when grafted onto a pre-trained ResNet50 backbone, as the superior model for this specific clinical domain. By constantly feeding high-resolution spatial feature maps from the early encoder layers through a dense grid of nested convolutions directly into the decoder, UNet++ achieved an

unparalleled Validation Intersection over Union (IoU) of 0.8605. This architecture demonstrated the highest robustness in preserving the absolute finest, single-pixel-wide capillary networks at the extreme, low-contrast periphery of the angiograms, effectively preventing the fragmentation that plagues simpler models.

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Mitigation of Extreme Class Imbalance

A fundamental technical hurdle addressed in this work was the severe class imbalance inherent in angiographic data, where vascular foreground pixels typically constitute less than 10% of the image matrix. The thesis demonstrated that optimizing neural networks solely on standard Binary Cross-Entropy results in catastrophic failure due to background domination.

This challenge was successfully mitigated through the theoretical formulation and empirical implementation of hybrid loss functions. By combining pixel-level optimization (BCE or Focal Loss) with regional overlap optimization (the differentiable soft Dice Loss), the gradient descent landscape was radically altered. The [Equation] formulation successfully forced the networks to prioritize intersection fidelity, entirely preventing background collapse and driving the high quantitative metrics observed during testing.

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Transition to Actionable Clinical Biomarkers

Perhaps the most significant contribution of this thesis is the integration of the mathematical skeletonization and topological analysis pipeline. Recognizing that a binary probability mask is merely an intermediate visual aid, the proposed methodology applied iterative thinning algorithms (such as the Zhang-Suen algorithm) to reduce the segmented vascular tree to a 1-pixel thick topological graph.

By programmatically traversing this graph, the system demonstrated the capability to autonomously, instantaneously, and objectively extract complex morphological counts---specifically the number of distinct connected vascular networks, the number of structural bifurcations (branch points), and the number of terminal capillaries (end points). These metrics, which are physically impossible for a clinician to extract manually at scale, provide direct, quantitative indicators of neovascularization,

ischemia, and overall vascular health.

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CAM Assay: Tumor Angiogenesis Assessment

The application of the complete pipeline to the CAM assay experiment yielded a clear and biologically significant finding: a biphasic dose-response in angiogenic activity. At the lowest concentration (0.1,[Equation]g), the compound exhibited a strongly pro-angiogenic profile---promoting vascular consolidation (network count decreasing from 13.38 to 4.00), massive branching (branch points reaching 5115), and functional loop closure (stable endpoint count despite growth). This pro-angiogenic response would, in a tumor microenvironment, create an ideal blood supply for cancer growth and metastasis.

At the highest concentration (10,[Equation]g), the compound exhibited the opposite, anti-angiogenic profile---causing vascular fragmentation (network count increasing from 12.00 to 18.00), collapse of branching complexity (branch points declining from 2765 to 1523), and accumulation of immature, non-functional vessel dead-ends (endpoints rising from 44 to 59). This anti-angiogenic response would effectively starve a tumor of blood supply. The intermediate dose (1,[Equation]g) showed a neutral, transitional response. This biphasic pattern is consistent with hormetic pharmacology and has direct implications for understanding the compound's therapeutic window in cancer-related applications.

Avenues for Future Research

While the methodologies developed in this thesis represent a significant advancement in automated angiography analysis, they also illuminate clear pathways for future academic research and clinical integration.

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Transition to 3D Volumetric Segmentation

The current pipeline operates on 2D planar projections of the vascular network. However, the human vascular system is inherently a complex 3D structure. The 2D projection inherently involves overlapping vessels, which complicates both the segmentation process and the downstream

topological counting. Future research must focus on transitioning the 2D UNet++ and DeepLabV3+ architectures into fully 3D volumetric convolution networks (e.g., 3D UNet++). Applying these models directly to 3D clinical imaging modalities, such as Magnetic Resonance Angiography (MRA) or Computed Tomography Angiography (CTA), would eliminate the overlap problem and allow for the extraction of true 3D spatial biomarkers.

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Adaptive Thresholding for Micro-Capillary Preservation

As identified in the limitations section of Chapter 6, the current post-processing pipeline relies on a static binarization threshold ([Equation]). While effective for primary vessels, this static threshold can inadvertently break the continuity of low-confidence micro-capillaries, generating false network fragmentations during skeletonization. Future work must investigate adaptive, hysteresis-based thresholding algorithms that track probability ridge-lines in the continuous output map, preserving the connectivity of delicate structures prior to the binary morphological thinning step.

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Real-time Clinical Deployment and Hardware Optimization

The massive parameter space of architectures like DeepLabV3+ currently requires substantial GPU acceleration, restricting their use to high-performance computing clusters. For this technology to be actively deployed in clinical operating rooms---for instance, to aid in real-time stent placement during fluoroscopy---the inference speed must be drastically reduced. Future research should explore network quantization, weight pruning, and knowledge distillation techniques to compress these heavy architectures into lightweight models capable of running in real-time on standard hospital hardware without sacrificing the structural fidelity of the vessel segmentations.

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Validation with Tumor Implantation on CAM

The current CAM assay analysis demonstrates dose-dependent angiogenic effects using morphological biomarkers extracted from the vascular skeleton. However, to definitively establish the anti-tumor efficacy of the high-dose concentration, future experiments should incorporate direct

tumor cell implantation onto the CAM membrane. By monitoring both tumor volume and vascular morphology simultaneously, researchers could directly correlate the anti-angiogenic effects observed in this thesis (vascular fragmentation, branch collapse, dead-end accumulation) with measurable tumor growth inhibition. Additionally, testing intermediate concentrations between 1,[Equation]g and 10,[Equation]g would help identify the minimum effective anti-angiogenic dose, and the incorporation of perfusion imaging (e.g., fluorescent dye injection) would validate whether the structural connectivity measured by skeletonization translates to actual blood flow.